



2022 TASK 1

Mathematics Extension 2

**General
Instructions**

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For all questions show relevant mathematical reasoning and/or calculations

**Total Marks:
50**

Section I – 5 marks

- Attempt Questions 1 – 3 on the Multiple Choice Answer sheet.

Section II – 45 marks

- Attempt Questions 6 – 8 in separate booklets.

Multiple Choice

5 Marks

Attempt Questions 1 – 5

Answer each question on your Multiple Choice Answer Sheet

1 What is the distance between the point $(3, -5, 2)$ and the z axis?

A. $\sqrt{13}$

B. $\sqrt{29}$

C. $\sqrt{34}$

D. $\sqrt{38}$

2 Let $\vec{OA} = -4\vec{i} - 3\vec{j} - 2\vec{k}$ and $\vec{OB} = \vec{i} - 5\vec{j} - \vec{k}$.

What the scalar projection of $\vec{OB}$ onto $\vec{OA}$?

A. $\frac{13}{3\sqrt{3}}$

B. $\frac{13}{\sqrt{29}}$

C. $\frac{13}{3\sqrt{87}}$

D. 13

3 Which of the following vectors lie within the plane $\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \left(\underline{p} - \begin{pmatrix} -5 \\ 0 \\ -1 \end{pmatrix} \right) = 0$?

A. $\begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$

B. $\begin{pmatrix} -4 \\ 3 \\ -6 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

D. $\begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}$

- 4 The vectors $\underline{m} = -3\underline{i} - 2\underline{j} + \lambda\underline{k}$ and $\underline{n} = 3\lambda\underline{i} + 10\underline{j} - 5\underline{k}$ are perpendicular.

What is the value of λ ?

A. -2

B. $-\frac{5}{7}$

C. $\frac{5}{7}$

D. 2

5 Consider the lines $\underline{l}_1 = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ and $\underline{l}_2 = \begin{pmatrix} 4 \\ -4 \\ 4 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ -3 \\ 9 \end{pmatrix}$.

Which of the following best describes the relationship between $\underline{l}_1$ and $\underline{l}_2$?

A. Parallel

B. Co-incident

C. Perpendicular

D. Orthogonal

End of Section I

Extended Response

45 Marks

Attempt Questions 6 – 8

Answer each question in a separate writing booklet.

In Questions 6 to 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks) Begin a new writing booklet.

(a) Let $\underline{a} = 5\underline{i} + 2\underline{j} - \underline{k}$ and $\underline{b} = -4\underline{i} - 4\underline{j} + 3\underline{k}$.

Find:

(i) $3\underline{a} - 4\underline{b}$ **1**

(ii) $|\underline{a}|$ **1**

(iii) $\hat{\underline{a}}$ **1**

(iv) The angle between $\underline{a}$ and $\underline{b}$. **2**

(v) The vector projection of $\underline{b}$ onto $\underline{a}$. **2**

(b) A parallelogram $OMPQ$ is created with the vectors $\overrightarrow{OP} = \underline{p}$ and $\overrightarrow{OQ} = \underline{q}$.

(i) Find the vector $\overrightarrow{OM}$ in terms of $\underline{p}$ and $\underline{q}$ **1**

(ii) It is known that $OMPQ$ is a rhombus. Prove that the diagonals of $OMPQ$ are perpendicular. **3**

- (c) (i) What can be said about the vectors $\underline{m}$ and $\underline{n}$ if the scalar projection of $\underline{m}$ onto $\underline{n}$ is the same as the scalar projection of $\underline{n}$ onto $\underline{m}$? Justify your answer. **2**
- (ii) What can be said about the vectors $\underline{m}$ and $\underline{n}$ if the **vector** projection of $\underline{m}$ onto $\underline{n}$ is the same as the **vector** projection of $\underline{n}$ onto $\underline{m}$? Justify your answer. **2**

End of Question 6

Question 7 (15 marks) Begin a new writing booklet.

- (a) The parallelogram $ABCD$ has vertices at the end points of the position vectors

$$\underline{a} = 2\underline{i} + 5\underline{j} - 4\underline{k}, \underline{b} = -\underline{i} - \underline{j} - 2\underline{k}, \underline{c} \text{ and } \underline{d} = -5\underline{i} + \underline{j} + 5\underline{k}$$

- (i) Find the vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$ 2

- (ii) Hence find the value of $\underline{c}$. 2

- (iii) Using vector methods, find the area of $ABCD$ 3

- (b) Two spheres, S_1 and S_2 , have equations $x^2 + y^2 + z^2 = 16$ and $(x - 3)^2 + y^2 + z^2 = 20$.

- (i) Find the value of x at which the two spheres meet. 1

- (ii) Hence find the equation of the circle of intersection between the two spheres. 2

- (c) Let the vector equation of a curve be $\underline{r}(t) = \begin{pmatrix} 2 \tan t \\ -1 \\ \cot 2t \end{pmatrix}$ 2

Find the Cartesian equation of the curve.

(d) Determine if the following vectors are linearly independent.

3

$$\underline{a} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$$

$$\underline{b} = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$$

$$\underline{c} = \begin{pmatrix} 8 \\ -3 \\ -2 \end{pmatrix}$$

End of Question 7

Question 8 (15 marks) Begin a new writing booklet.

(a) Let the three points A , B and C represent the points $(3, -2, -1)$, $(-5, 1, 1)$ and $(9, -4, -1)$.

(i) The vector equation of the line AB .

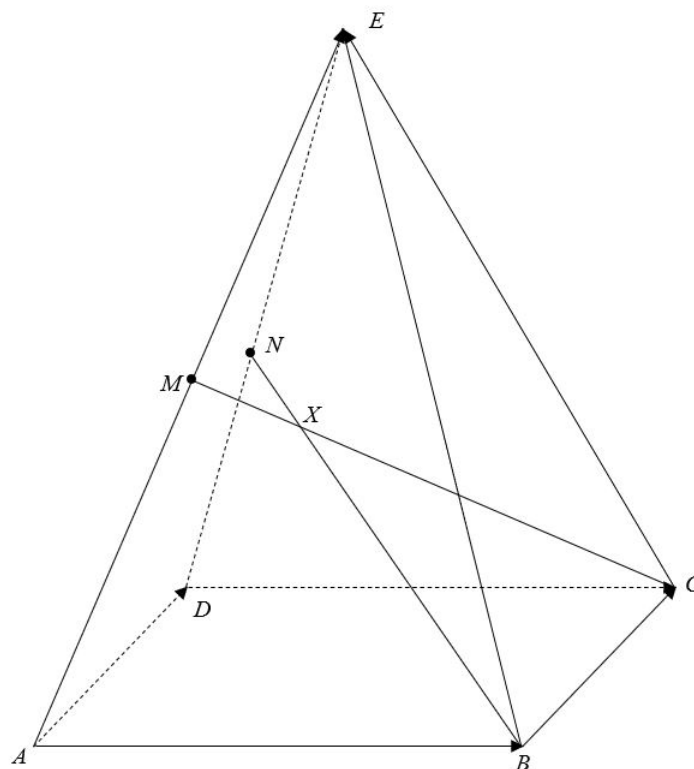
3

(ii) Find the minimum distance between the line AB and the point C .

3

(b) The solid $ABCDE$ is a right rectangular pyramid with the point E as the apex. The mid-points of AE and DE are M and N respectively. It is known that the lines MC and NB meet at the point X .

Let $\vec{AB} = \underline{p}$, $\vec{AD} = \underline{q}$ and $\vec{AE} = \underline{r}$.



Question 8 continues on page 10

Question 8 (continued)

- (i) Find the vector $\overrightarrow{DE}$ in terms of $\underline{p}$, $\underline{q}$ and/or $\underline{r}$. **1**
- (ii) Find the vectors $\overrightarrow{NB}$ and $\overrightarrow{MC}$ in terms of $\underline{p}$, $\underline{q}$ and/or $\underline{r}$. **2**
- (iii) Hence or otherwise show that $\frac{|\overrightarrow{MX}|}{|\overrightarrow{XC}|} = \frac{|\overrightarrow{NX}|}{|\overrightarrow{XB}|} = \frac{1}{2}$. **3**

The solid is now placed on a three dimensional Cartesian plane such that A is on the origin,
 $B = (4, 0, 0)$, $D = (0, 2, 0)$ and $E = (2, 1, 5)$.

- (iv) Find the smallest angle between the vector $\overrightarrow{EX}$ and the plane containing $\triangle XBC$. **3**

End of Question 8

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

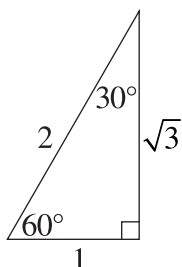
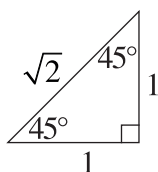
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

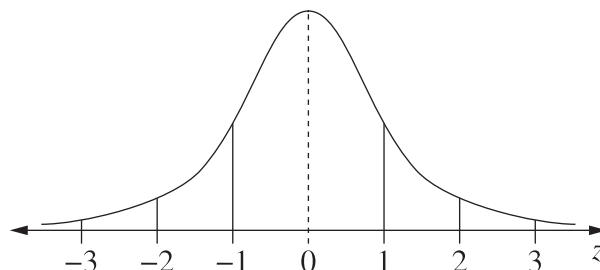
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

12 Ext 2 2023 Task 1 Solutions RDS - MC

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Q1 Distance from z axis = $\sqrt{x^2 + y^2}$
= $\sqrt{9+25}$
= $\sqrt{34} \therefore \textcircled{C}$

Q2 $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$
= $\frac{\vec{OB} \cdot \vec{OA}}{|\vec{OA}|}$
= $\frac{(-4)(1) + (-3)(-5) + (-2)(-1)}{\sqrt{4^2 + 3^2 + 2^2}}$
= $\frac{-4 + 15 + 2}{\sqrt{29}}$
= $\frac{13}{\sqrt{29}} \therefore \textcircled{B}$

Q3 $\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \left(\vec{p} - \begin{pmatrix} -5 \\ 0 \\ -1 \end{pmatrix} \right) = 0$

let $\vec{p} = (x, y, z)$

$$2(x+5) + 1(y) + 1(z+1) = 0$$

$$2x+10 + y + z + 1 = 0$$

$$2x+y+z = -11$$

let $x=-4, y=3, z=-6$

$$-8 + 3 - 6 = -11 \therefore \textcircled{B}$$

Q4 $\vec{m} \cdot \vec{n} = 0$

$$-3 \times 3\lambda - 2 \times 10 + \lambda \times -5 = 0$$

$$-9\lambda - 20 - 5\lambda = 0$$

$$-14\lambda = 20$$

$$\lambda = -\frac{10}{7} \therefore \textcircled{B}$$

Q5 $\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \neq k \begin{pmatrix} 3 \\ -3 \\ 9 \end{pmatrix} \therefore \text{Not parallel}$

$$\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 9 \end{pmatrix} = -3 - 3 + 27 \neq 0 \therefore \text{Not perpendicular}$$

Solve $\begin{cases} 2 - \lambda_1 = 4 + 3\lambda_2 & \textcircled{1} \\ -1 + \lambda_1 = -4 - 3\lambda_2 & \textcircled{2} \end{cases}$ Solve

$$2 + 3\lambda_1 = 4 + 9\lambda_2$$

$$\textcircled{1} + \textcircled{2}$$

$$1 = 0 \quad \text{Not possible} \therefore \textcircled{D}$$

Answers :

Q1	Q2	Q3	Q4	Q5
C	B	B	B	D

12 Ext 2 2022 Task 1 Solutions RDS - Q6

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$$(a) (i) \quad 3\underline{a} - 4\underline{b} = 3 \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} - 4 \begin{bmatrix} -4 \\ -4 \\ 3 \end{bmatrix} \\ = 31\underline{i} + 22\underline{j} - 15\underline{k}$$

$$(ii) \quad |\underline{a}| = \sqrt{5^2 + 2^2 + 1^2} \\ = \sqrt{30}$$

$$(iii) \quad \hat{\underline{a}} = \frac{1}{\sqrt{30}} \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

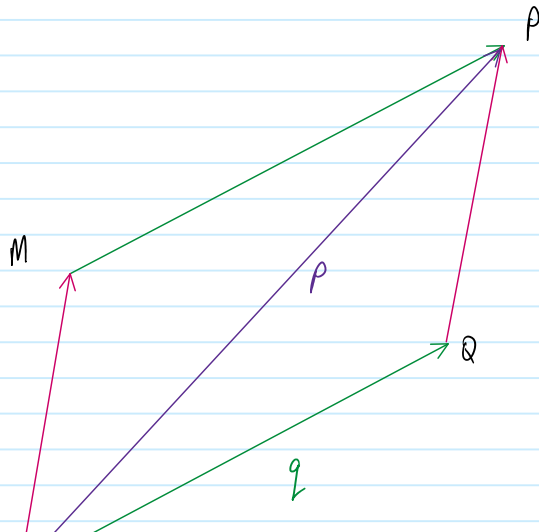
$$(iv) \quad \cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} \quad \underline{a} \cdot \underline{b} = -20 - 8 - 3 = -31 \\ \theta = \cos^{-1} \left[\frac{-31}{\sqrt{30} \times \sqrt{41}} \right] \quad |\underline{a}| = \sqrt{30} \quad |\underline{b}| = \sqrt{41}$$

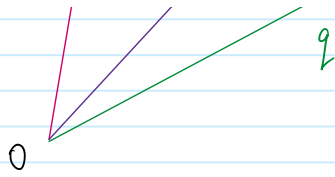
$$= 27.862 \dots$$

$$= 28^\circ \text{ (Nearest degree)} \quad (1)$$

$$(v) \quad \text{Proj}_{\underline{a}} \underline{b} = \left[\frac{\underline{a} \cdot \underline{b}}{|\underline{a}|^2} \right] \times \underline{a} \quad (1) \text{ scalar projection} \\ = \left[\frac{-31}{30} \right] \times \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} \quad (1)$$

(b)





(i) $\vec{OM} + \vec{MP} = \vec{OP}$ $\vec{MP} = q$ (opposite sides of a parallelogram)
 $\therefore \vec{OM} = p - q$ (1)

(ii) RTP $\vec{MQ} \cdot \vec{OP} = 0$ (1)
 let $\vec{OM} = m$

If OMPQ is a Rhombus then $|q| = |m|$

$\therefore q \cdot q = m \cdot m$

$\vec{MQ} = q - m$
 $\vec{OP} = m + q$ (1)

$\therefore \vec{MQ} \cdot \vec{OP} = (q - m) \cdot (q + m)$
 $= q \cdot q - \cancel{q \cdot m} + \cancel{q \cdot m} - m \cdot m$
 $= |q|^2 - |m|^2$
 $= 0 \quad (\text{As } |q| = |m|)$ (1)

$\therefore$ Diagonals are perpendicular.

(c) (i) The vectors m & n must be the same length or perpendicular

$\frac{m \cdot n}{|m|} = \frac{m \cdot n}{|n|}$ if $\frac{m \cdot n}{\text{projection}} = 0$ then both are zero.

$\therefore |m| = |n|$ (1)

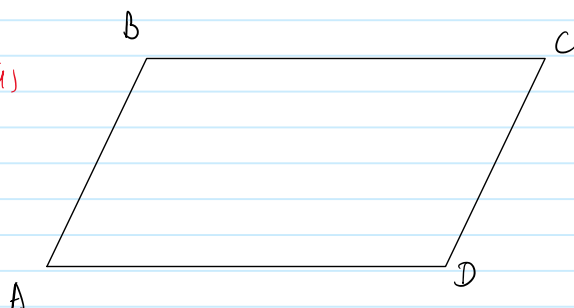
(ii) The vectors m & n must be the same vector or they are perpendicular. (1)

12 Ext 2 2023 Task 1 Solutions RDS - Q7

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(a)

(i)



$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \begin{pmatrix} -1-2 \\ -1-5 \\ -2+4 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ -6 \\ 2 \end{pmatrix} \quad (1)$$

$$\vec{AD} = \vec{OD} - \vec{OA}$$

$$= \begin{pmatrix} -5-2 \\ 1-5 \\ 5+4 \end{pmatrix}$$

$$= \begin{pmatrix} -7 \\ -4 \\ 9 \end{pmatrix} \quad (1)$$

(ii) $\vec{AC} = \vec{AB} + \vec{BC}$

($\vec{BC} = \vec{AD}$ As ABCD is a parallelogram)

$$\therefore \vec{AC} = \begin{pmatrix} -10 \\ -10 \\ 11 \end{pmatrix} \quad (1)$$

$$\vec{OC} = \vec{OA} + \vec{AC}$$

$$\therefore \vec{OC} = \begin{pmatrix} 2-10 \\ 5-10 \\ -4+11 \end{pmatrix}$$

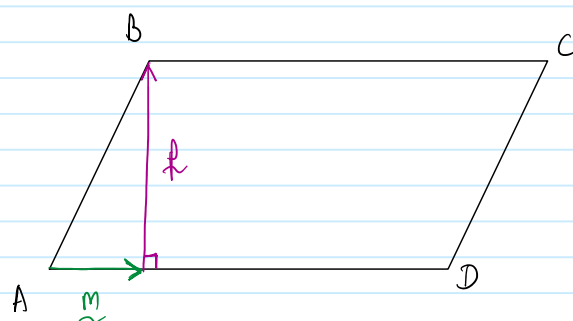
$$= \begin{pmatrix} -8 \\ -5 \\ 7 \end{pmatrix} \quad (1)$$

(iii) Scalar projection of $\vec{AB}$

onto $\vec{AD} = |\underline{m}|$

$$|\underline{m}| = \frac{\vec{AB} \cdot \vec{AD}}{|\vec{AD}|}$$

$$= \frac{21+24+16}{\sqrt{146}}$$



$$= \frac{63}{\sqrt{146}} \quad (1)$$

$$\begin{aligned} \therefore |p|^2 &= |AB|^2 - |m|^2 \\ &= 49 - \frac{63^2}{146} \\ &= \frac{3185}{146} \end{aligned}$$

$$\therefore |p| = \sqrt{\frac{3185}{146}} \quad (1)$$

$$\therefore \text{Area of } ABCD = \sqrt{3185}$$

$$\doteq 56.4358 \dots$$

$$= 56.44 \quad (2dp) \quad (1)$$

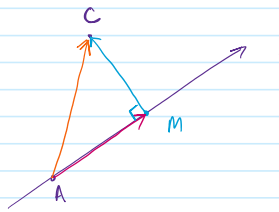
12 Ext 2 2023 Task 1 Solutions RDS - Q8

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(a) (i) $\vec{AB} = \begin{pmatrix} -5-3 \\ 1-(-2) \\ 1-(-1) \end{pmatrix}$
 $= \begin{pmatrix} -8 \\ 3 \\ 2 \end{pmatrix}$

let line l go through A & B

$\therefore l = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 3 \\ 2 \end{pmatrix}$ Any multiple of this.
 (1) Can be A or B (1)

(ii)  $\vec{AB} = \begin{pmatrix} -8 \\ 3 \\ 2 \end{pmatrix}$ $|\vec{AB}| = \sqrt{77}$
 $\vec{AC} = \begin{pmatrix} 9-3 \\ -4-(-2) \\ -1-(-1) \end{pmatrix} = \begin{pmatrix} 6 \\ -2 \\ 0 \end{pmatrix}$ $|\vec{AC}| = \sqrt{40}$ (1)

let M be on the line l so that $\angle AMC = 90^\circ$

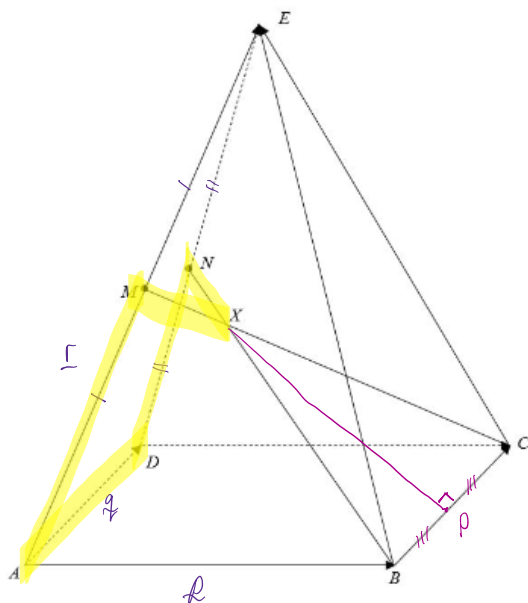
$\therefore |\vec{AM}| = \text{Proj}_{\vec{AB}} \vec{AC} = \frac{\vec{AC} \cdot \vec{AB}}{|\vec{AB}|}$
 $= \frac{-8 \times 6 + 3 \times -2 + 0}{\sqrt{77}}$ (1)
 $= \frac{-54}{\sqrt{77}}$

$|\vec{MC}| = \sqrt{|\vec{AC}|^2 - |\vec{AM}|^2}$
 $= \sqrt{40 - \frac{2916}{77}}$

$= 1.4594...$

$= 1.46$ (2dp) (1)

(b)



(i) $\vec{AD} + \vec{DE} = \vec{AE}$
 $\therefore \vec{DE} = \vec{l} - \vec{q}$

(ii) $\vec{DN} = \frac{1}{2} \vec{DE}$
 $= \frac{1}{2} (\vec{l} - \vec{q})$

$$\vec{AM} = \frac{1}{2} \vec{AE}$$
$$= \frac{1}{2} (\vec{l})$$

$$\vec{AD} + \vec{DN} + \vec{NB} + \vec{BA} = \vec{0}$$
$$\vec{NB} = \vec{AB} - \vec{AD} - \vec{DN}$$
$$= \vec{r} - \vec{q} - \frac{1}{2} (\vec{l} - \vec{q})$$
$$= \vec{r} - \frac{1}{2} (\vec{q} + \vec{l}) \quad (1)$$

$$\vec{AM} + \vec{MC} + \vec{CB} + \vec{BA} = \vec{0}$$
$$\vec{MC} = \vec{AB} + \vec{BC} - \vec{AM}$$
$$= \vec{r} + \vec{q} - \frac{1}{2} \vec{l} \quad (1)$$

(iii) let $\vec{MX} = \lambda \vec{MC}$ & $\vec{NX} = \sigma \vec{NB} \quad (1)$

from **highlight** $\vec{AD} + \vec{DN} + \vec{NX} + \vec{XM} + \vec{MA} = \vec{0}$

$$\vec{q} + \frac{1}{2} (\vec{l} - \vec{q}) + \sigma (\vec{r} - \frac{1}{2} \vec{q} - \frac{1}{2} \vec{l}) - \lambda (\vec{r} + \vec{q} - \frac{1}{2} \vec{l}) - \frac{1}{2} \vec{l} = \vec{0}$$

$$\vec{q} (1 - \frac{1}{2} - \frac{\sigma}{2} - \lambda) + \vec{r} (\sigma - \lambda) + \vec{l} (\frac{1}{2} - \frac{\sigma}{2} + \frac{\lambda}{2} - \frac{1}{2}) = \vec{0}$$

All coefficients must be zero

$$\therefore -\frac{\sigma}{2} - \lambda = -\frac{1}{2}$$

$$\sigma = \lambda$$

$$\frac{\sigma}{2} = \frac{\lambda}{2}$$

$$\sigma - \lambda = 0$$

$$-\frac{\sigma}{2} - \sigma = -\frac{1}{2}$$

$$-\frac{3\sigma}{2} = -\frac{1}{2}$$

$$\sigma = \frac{1}{3} \quad \therefore \lambda = \frac{1}{3}$$

$$\therefore \vec{MX} = \frac{1}{3} \vec{MC} \quad \therefore \vec{XC} = \frac{2}{3} \vec{MC}$$

$$\vec{NX} = \frac{1}{3} \vec{NB} \quad \therefore \vec{XB} = \frac{2}{3} \vec{NB}$$

$$\therefore \frac{\vec{MX}}{\vec{XC}} = \frac{\frac{1}{3} \vec{MC}}{\frac{2}{3} \vec{MC}} = \frac{1}{2} \quad \frac{\vec{NX}}{\vec{XB}} = \frac{\frac{1}{3} \vec{NB}}{\frac{2}{3} \vec{NB}} = \frac{1}{2}$$

(1)

$$(w) \quad p = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} \quad q = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \quad r = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$$

$$\begin{aligned} \vec{AX} &= \vec{AM} + \vec{MX} \\ &= \frac{1}{2} \vec{AE} + \frac{1}{3} \vec{MC} \\ &= \frac{1}{2} \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} + \frac{1}{3} \left(\begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \right) \\ &= \begin{pmatrix} 1 \\ \frac{1}{2} \\ \frac{5}{2} \end{pmatrix} + \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ -\frac{5}{6} \end{pmatrix} \\ &= \begin{pmatrix} \frac{4}{3} \\ 1 \\ \frac{5}{3} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \vec{AE} &= \vec{AX} + \vec{XE} \\ \therefore \vec{XE} &= \vec{AE} - \vec{AX} \\ &= \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} - \begin{pmatrix} \frac{4}{3} \\ 1 \\ \frac{5}{3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{2}{3} \\ 0 \\ \frac{10}{3} \end{pmatrix} = \frac{2}{3} \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} \quad (1) \quad |\vec{XE}| = \frac{2\sqrt{26}}{3} \end{aligned}$$

let P be the midpoint of BC

$$\therefore \vec{AP} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{AX} = \vec{AP} + \vec{PX}$$

$$\therefore \vec{PX} = \vec{AX} - \vec{AP}$$

$$\vec{AX} = \vec{AP} + \vec{PX}$$

$$\therefore \vec{PX} = \vec{AX} - \vec{AP}$$

$$= \begin{pmatrix} \frac{4}{3} \\ 1 \\ \frac{5}{3} \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{8}{3} \\ 0 \\ \frac{5}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -8 \\ 0 \\ 5 \end{pmatrix} \quad \therefore |\vec{PX}| = \frac{\sqrt{89}}{3}$$

①

or finding the
plane containing
 $\triangle XBC$

Angle between $\vec{PX}$ & $\vec{EX}$

$$\theta = \cos^{-1} \left[\frac{|\vec{PX} \cdot \vec{EX}|}{|\vec{PX}| |\vec{EX}|} \right]$$

$$\vec{PX} \cdot \vec{EX} = \frac{2}{3} \times \frac{1}{3} \left[-8 + 0 + 25 \right]$$

$$= \frac{34}{9}$$

$$\therefore \theta = \cos^{-1} \left[\frac{\frac{34}{9}}{\frac{\sqrt{89}}{3} \times \frac{2\sqrt{6}}{3}} \right]$$

$$= 69.3045 \dots$$

$$= 69^\circ \text{ (Nearest degree)} \quad \textcircled{1}$$